KAIZEN MATHS

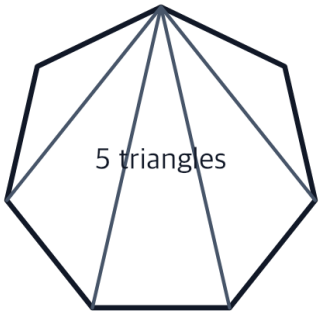
Lesson starter

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

SA1



Find the sum of the interior angles of a heptagon.

GEO-BAE7-L1 · SA

Show your working

KAIZEN MATHS

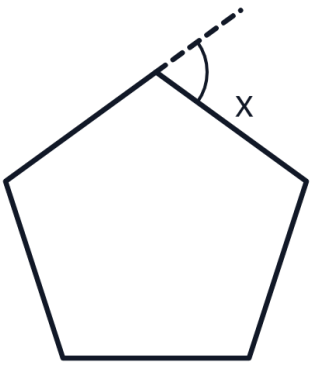
Lesson starter

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

SB1



Find the sum of the exterior angles of this pentagon.

GEO-BAE7-L1 · SB

Show your working

KAIZEN MATHS

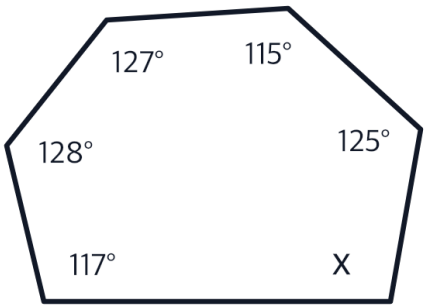
Lesson starter

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

SC1



Find the missing interior angle x.

GEO-BAE7-L1 · SC

Show your working

KAIZEN MATHS

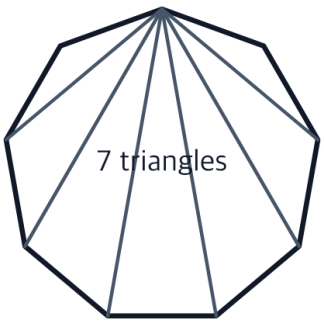
Lesson starter

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

SD1



Find the sum of the interior angles of a nonagon.

GEO-BAE7-L1 · SD

Show your working

KAIZEN MATHS

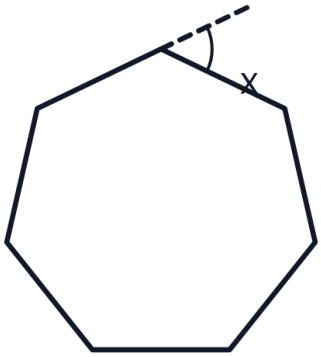
Exit ticket

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

EA1



Find the sum of the exterior angles of this heptagon.

GEO-BAE7-L1 · EA

Show your working

KAIZEN MATHS

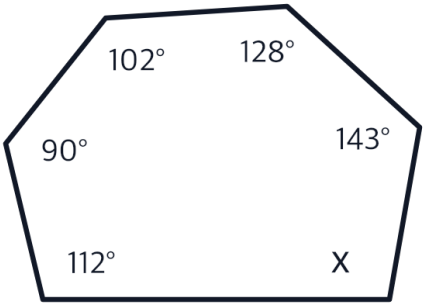
Exit ticket

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

EB1



Find the missing interior angle x.

GEO-BAE7-L1 · EB

Show your working

KAIZEN MATHS

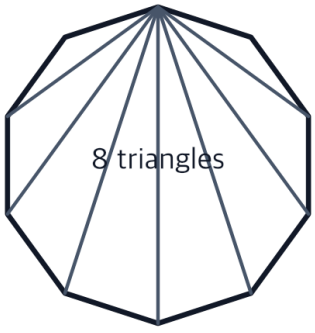
Exit ticket

Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

EC1



Find the sum of the interior angles of a decagon.

GEO-BAE7-L1 · EC

Show your working

KAIZEN MATHS

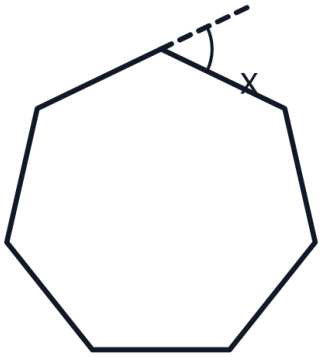
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Polygons and Interior/Exterior Angles

Core Angle Facts

Name _____

EA1



Find the sum of the exterior angles of this heptagon.

GEO-BAE7-L1 · EA

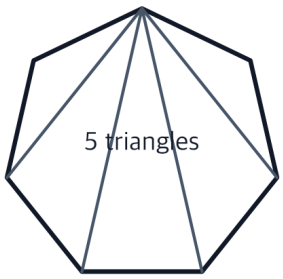
Show your working

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Find the sum of the interior angles of a heptagon.

Answer:

900°

Working:

A heptagon has 7 sides.

A polygon with n sides can be split into n - 2 triangles from one vertex.

Number of triangles = 7 - 2 = 5.

Interior angle sum = (7 - 2) × 180°.

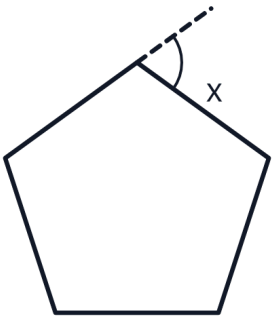
Interior angle sum = 5 × 180° = 900°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Find the sum of the exterior angles of this pentagon.

Answer:

360°

Working:

Walking around any convex polygon makes one complete turn.

One complete turn is 360°.

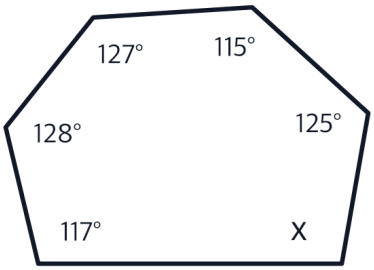
So the exterior angles add to 360°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Find the missing interior angle x.

Answer:

108°

Working:

This is a hexagon, so it has 6 sides.

Interior angle sum = (6 - 2) × 180° = 720°.

Known angles = 117° + 128° + 127° + 115° + 125° = 612°.

x = 720° - 612°.

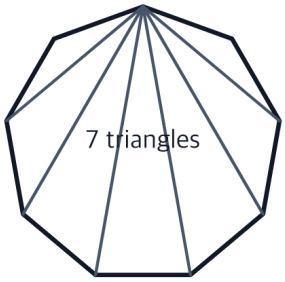
x = 108°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Find the sum of the interior angles of a nonagon.

Answer:

1260°

Working:

A nonagon has 9 sides.

A polygon with n sides can be split into n - 2 triangles from one vertex.

Number of triangles = 9 - 2 = 7.

Interior angle sum = (9 - 2) × 180°.

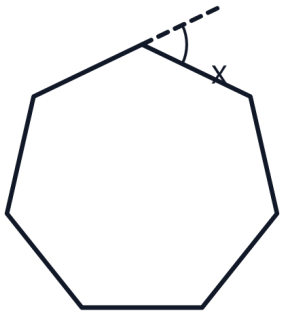
Interior angle sum = 7 × 180° = 1260°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Find the sum of the exterior angles of this heptagon.

Answer:

360°

Working:

Walking around any convex polygon makes one complete turn.

One complete turn is 360°.

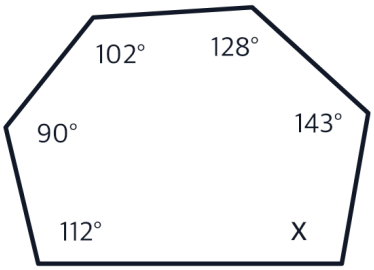
So the exterior angles add to 360°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Find the missing interior angle x.

Answer:

145°

Working:

This is a hexagon, so it has 6 sides.

Interior angle sum = (6 - 2) × 180° = 720°.

Known angles = 112° + 90° + 102° + 128° + 143° = 575°.

x = 720° - 575°.

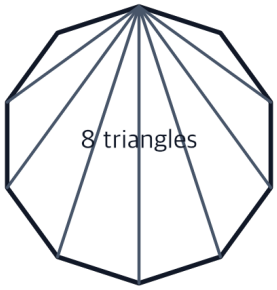
x = 145°.

Teacher answers & working

Polygons and Interior/Exterior Angles · Core Angle Facts

GEO-BAE7-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Find the sum of the interior angles of a decagon.

Answer:

1440°

Working:

A decagon has 10 sides.

A polygon with n sides can be split into n - 2 triangles from one vertex.

Number of triangles = 10 - 2 = 8.

Interior angle sum = (10 - 2) × 180°.

Interior angle sum = 8 × 180° = 1440°.